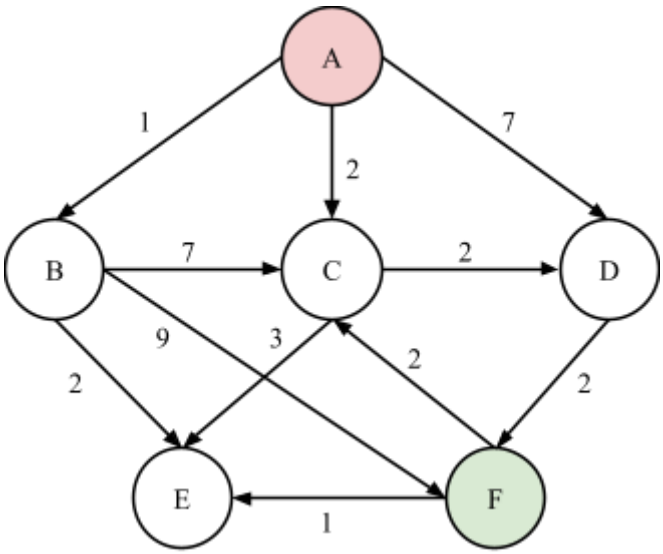
AI 500 - Foundations of AI
Quiz 1 - Search - 20th September 2025

Name:_____ Roll Number:_____

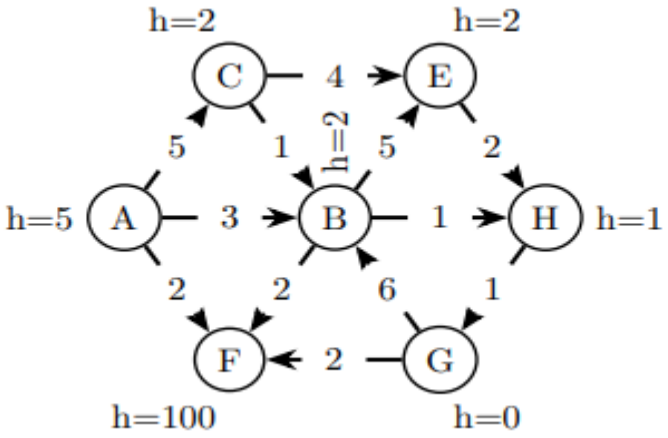
Q1. For the weighted graph below with specified edge costs, use UCS (uniform cost search) to determine the lowest cost path *g(n)* from the start node, **A**, to the goal node, **F**.



In the space provided below, clearly

- Show the **frontier** (priority queue) and **explored** state at each step of the expansion. (**4 marks**)
- List the **final path** found and state its **total cost**. (**2 marks**)

Q2. Given the search problem defined by the following graph, where **A** is the initial state and **G** is the goal state and the cost of each action is indicated on the corresponding edge. Cost function => f(n)=g(n)+h(n)



(a) Construct the complete **labelled search tree** for the **A* algorithm** from initial state A to goal state G using the given directed graph **(3 marks)**

(b) Determine the optimal path and its total cost (**1 mark)**

Q1 Solution)

Question 1 Solution

Final path: A → C → D → F
Total cost: 2+2+2=6 } 2 marks for this

Step 0 : Frontier: (B,1), (C,2), (D,7)
Explored: (A,0)

Step 1 : Frontier: (C,2), (E,3), (D,7), (F,10)
Explored: (A,0), (B,1)

Step 2: Frontier: (E,3), (D,4), (F,10)
Explored: (A,0), (B,1), (C,2)

Step 3: Frontier: (D,4), (F,10)
Explored: (A,0), (B,1), (C,2), (E,3)

Step 4: Frontier: (F,6)
Explored: (A,0), (B,1), (C,2), (E,3) (D,4)

Step 5 : Frontier :
Explored : (A,0), (B,1), (C,2), (E,3) (D,4), (F,6)

} 4 marks for this

Q2 Solution)

